Acta Astronautica 189 (2021) 10–25

Contents lists available at ScienceDirect

Acta Astronautica

journal homepage: www.elsevier.com/locate/actaastro

Overview of reusable space systems with a look to technology aspects

Paolo Baiocco

CNES, 2 Place Maurice-Quentin, Paris, 75001, France

ARTICLE INFO

Keywords:
Space systems
Recovery
Reusability
Tehnology
Re-entry

ABSTRACT

Rocket reusability dates as long as rockets. Since the beginning of space age, the possibility to reuse the full vehicle or some stages has been a matter of engineering. But this was directly opposite to mass ratio efficiency, the key for a well-designed rocket. Since as soon as a stage has to be recovered, subsystems to provide this function and additional inert mass are needed. Even if reusable systems are only a small part of the worldwide rockets, they flew operationally since the space shuttle age. However, full reusable vehicles are not yet operational and only partially reusable launch vehicles have been flown until now. The preliminary space shuttle design consisted of two stages both liquid and reusable (booster and orbiter stages). The consolidated design retrieved the orbiter (upper stage, the cargo and propulsion bay) and the solid rocket boosters. On Russian side Energia-Buran only retrieved the upper stage with its cargo bay and the same approach was adapted for the original design of Ariane 5-Hermes. Europe investigated and developed different reusable systems, set up test facilities and flew the IXV demonstrator in 2015. More recently, the preferred strategy consists in reuse of unwinged booster stages (e.g. Falcon 9 of SpaceX) or of only the orbited spacecraft (e.g. NASA-USAF X-37). France and Europe are preparing propulsion and reusable rocket stage demonstration bricks for next generation launchers. The purpose of this paper is to provide an overview of some reusable space systems with emphasis on the different design strategy and skills mainly focused on European heritage. The systems are split into low and high energy with a view to technology aspects and in-flight experiments.

1. Introduction

Recent developments of Falcon 9 [1] and Starship took to a regain of interest for rocket reusability. The idea to reuse rockets, entirely or partially, dates back to the first Apollo designs by Von Braun implementing winged stages. Finally, only the Apollo capsule was retrievable, but not reusable. The reusability came back with space shuttle. This was initially conceived as a fully reusable two stage liquid rocket, but the final design opted for a partially reusable system with solid rocket boosters.

A similar design was adopted by the Soviet/Russian Energia-Buran, which performed the successful maiden flight in unmanned mode in 1988 [2]. This was also the last flight of this system using only liquid propulsion and retrieving the Buran shuttle with its AOCS (Attitude and Orbit Control System) module, the main propulsion being on the core stage.

Hermes [3] design principles were similar to those of Energia-Buran, being the recovered part only the manned spacecraft. European decision to abandon manned flight resulted in keeping only the cargo version of Ariane 5, but winged bodies reusable technologies had been developed and test benches built for this program. This industrial setup allowed Europe to develop later on the ARD and Pre-X/IXV experimental re-entry vehicles [4] as well as technology spin-off (e.g. carbon sic body flaps, carbon-carbon nozzles).

A number of projects and associated technology developments allowed to progress in the frame of reusable vehicles mostly for orbit re-entry. Thus, for example, the filament wound tanks cryogenic technology progressed thanks to the X-33 program and the reusable technology heritage allowed to develop the operational X-37 for Low Earth Orbit (LEO) operations.

On the Soviet/Russian side, the technology and the demonstrators developed for Buran allowed the current development of the Dream Chaser and the use of the very performing wind tunnels of TSAGI.

But the last innovation came with the reuse of low energy stages, i.e. the rocket boosters, flown by SpaceX on the principle developed by McDonnell Douglas [5]. Experience in the frame of vertical-lift-off/vertical landing was already gained in the frame of DC-X [6] and Kistler [7] programs, respectively in the frame of subsonic maneuverability/rapid turnaround and subsystems for a two stage fully reusable vehicle.

On Russian side, quite an effort had been dedicated to the

E-mail address: paolo.baiocco@cnes.fr.

https://doi.org/10.1016/j.actaastro.2021.07.039
Received 31 March 2021; Received in revised form 2 July 2021; Accepted 27 July 2021
Available online 4 August 2021

Acronyms & symbols

Acronyms

AoA	Angle of Attack
AOCS	Attitude and Orbit Control System
ARD	Atmospheric Re-entry Demonstrator
ATD	Aero Thermo Dynamics
ATV	Automated Transfer Vehicle
CFD	Computational Fluid Dynamics
CMC	Carbon Matrix Composite
COTS	Components Off-The Shelf
EAP	Etage à Poudre
EGSE	Electrical GSE
ELV	Expendable Launch Vehicle
FCS	Flight Control System
FPA	Flight Path Angle
FSL	Sea level thrust
GLOW	Gross Lift-Off Weight
GNC	Guidance, Navigation and Control
GNSS	Global Navigation Satellite System
GSE	Ground Support Equipment
GTO	Geostationary Transfer Orbit
HMS	Health Monitoring System
IMU	Inertial Measurement Unit
IRDT	Inflatable Re-entry and Descent Technology
IRL	Integration Readiness Level
Lift/Drag	Lift over Drag
LEO	Low Earth Orbit
LH2	Liquid Hydrogen
LOX	Liquid Oxygen
LV	Launch Vehicle
MECO	Main Engine Cut-Off
MGSE	Mechanical GSE
MRO	Maintenance Repair & Overhaul
NRC	Non-Recurring Cost
PL	Pay Load
RC	Recurring Cost
RCS	Reaction Control System
RLV	Reusable Launch Vehicle
RSR	Reusable Sounding Rocket
RRSD	Reusable Rocket Stage Demonstration
R&T	Research and Technology
SHM	Structural Health Monitoring
SRB	Solid Rocket Boosters
S/S	Sub System
SSME	Space Shuttle Main Engine
TAEM	Terminal Area Energy Management
TPS	Thermal Protection System
TRL	Technology Readiness Level
TSTO	Two Stage To Orbit
TVC	Thrust Vector Control
VTVL	Vertical Take-off Vertical Landing
WRT	With Respect To
WTT	Wind Tunnel Test

SYMBOLS, Latin

I_{sp}	Specific Impulse
K	Ratio of experimental vehicle atmospheric density to reference vehicle one
L	Characteristic Length
m	mass
M	Mach number
Re	Reynolds number
R_{Nose}	Nose Radius
V	Relative velocity

Greek

ΔV_{id}	Ideal velocity increment
ε	stage mass ratio
λ	Ratio of experimental vehicle length to reference vehicle one
μ	Dynamic viscosity
π	mass fraction
ρ	Atmospheric density
Φ	Heat flux

preliminary design of reusable winged boosters thorough the Baikal project, proposed as alternative to the expendable boosters on the launcher Angara [8]. On this same principle the CNES/TSNIIMASH cooperation investigated the Bargouzin concept, a winged couple of fly-back boosters for Ariane 5 launcher [9].

Nowadays the majority of launch systems is expendable and few reusable systems are currently operational (e.g. Falcon 9 and X–37B), but demonstration programs and developments are increasing worldwide showing a renewed interest for reusability. The reason for this trend is based on the success of SpaceX and on the following considerations:

- Reusability can provide an economic advantage depending on the launch rate, the recovery/refurbishment cost and the filling ratio of the mission (ratio between the actual and maximum payload mass).
- Reuse of stages can avoid fallout in sea or lands. The consequences can be more or less important depending on parameters like the population density in fallout zones and the potential pollution of stages components in these areas.

The two above points are not developed in this paper, which is focused on the survey of a limited number of reusable vehicles. Different families of reusable vehicles can be defined depending on the energy associated, i.e. the corresponding sum of the kinetic and potential energies. These categories can be split into three main families (Fig. 1): rockets, orbit services and suborbital vehicles. In the following, focus will be given mainly to rockets and orbit services. The survey of reusable systems will be split into two domains: low and high energy, where low energy refers to booster stages and high energy to orbited stages or spacecraft. In view of the raising interest in reusable systems, it is useful to analyze some of the space systems investigated nowadays and in the past with the related advantages and disadvantages as well as the corresponding technology challenges.

The following of the paper is structured as follows:

- A review of the low energy space systems and corresponding skills: boosters' stages and related demonstrators.
- A review of high energy space systems and corresponding skills: winged/lifting bodies, capsules and related demonstrators.
- Some technology aspects of both low and high energy space systems.

The corresponding recovery strategies are outlined.

2. Low energy stage recovery and reuse

As it was shown in Refs. [10,11], many options to recover booster stages exist depending on the stages features (e.g. winged or unwinged). They present advantages and disadvantages as well as different critical

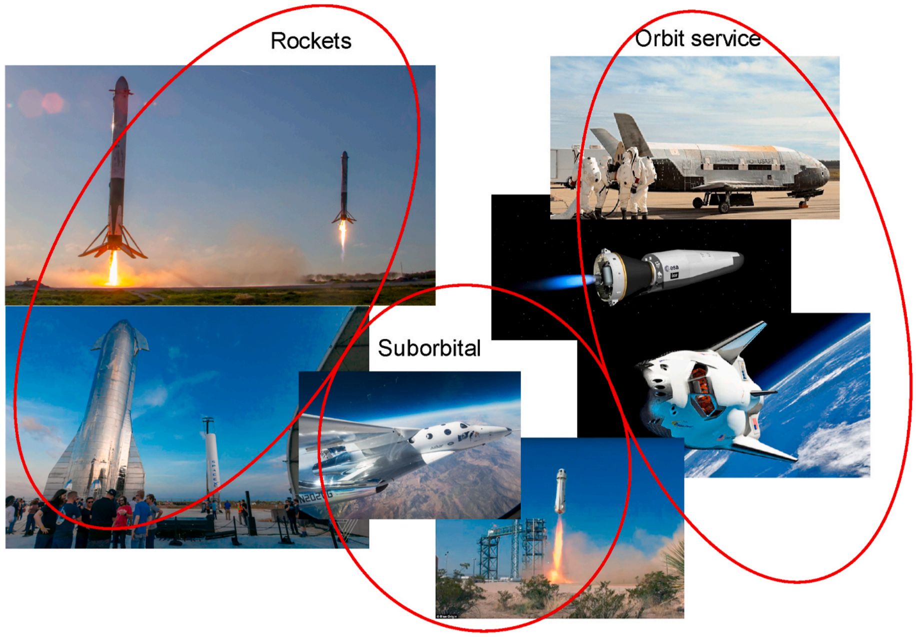


Fig. 1. Examples of new generation reusable spacecraft.

technology. They are distinguished into (Fig. 2):

A. Toss-back (also called boost-back), unwinged. This option consists in recovering the booster stage on the launch site after maneuvering using main propulsion (e.g. SpaceX strategy).
B. Barge landing, unwinged. This option consists in booster stage landing on a barge maneuvering during ballistic trajectory.
C. Gliding back, winged (dead-leaf and glide-back maneuver). This option consists in recovering the booster stage on the launch site after maneuvering using main propulsion and then gliding back to launch site.
D. Ballistic flight and barge landing, unwinged, this option consists in recovering the booster stage on a barge after ballistic flight.
E. Ballistic flight, rocket boost and glide back, winged. This option consists in recovering the booster stage maneuvering after ballistic flight using main rocket propulsion and then a ballistic flight back to launch site.
F. Ballistic flight and cruising back, winged. This option consists in rocket booster trimmed flight to launch site after maneuvering using aircraft like features.

The option choice depends on many parameters, such as the separation Mach number, the mass/centering of the stage, the recovery means and the refurbishment phase. Different options imply different critical technologies. For instance, a booster stage can be recovered by means of parachute after the ballistic phase (as it was the case of the space shuttle Solid Rocket Boosters - SRB and of the Ariane 5 Etage à poudre - EAP) and a small stage (or vehicle) by means of paraglider. Toss-back and barge-landing require the mastering of multiple engine ignitions and most of landing solutions imply precise guidance [12].

In the following it will be shown how some of these strategies have been implemented and which are the corresponding main technology aspects.

Since the beginning of space programs, booster stage recovery has been considered for different reasons, such as expertise and reuse for other missions. The most famous example is the recovery system investigation for the Saturn IB booster. A method for gliding back was proposed at NASA Langley based on a paraglider to glide back the stage to the launch site [13]. The paraglider is able to control the re-entry forces, provide necessary lateral turns, and land the booster at a very slow descent rate. The paraglider was tested on Wind Tunnel (WTT) in

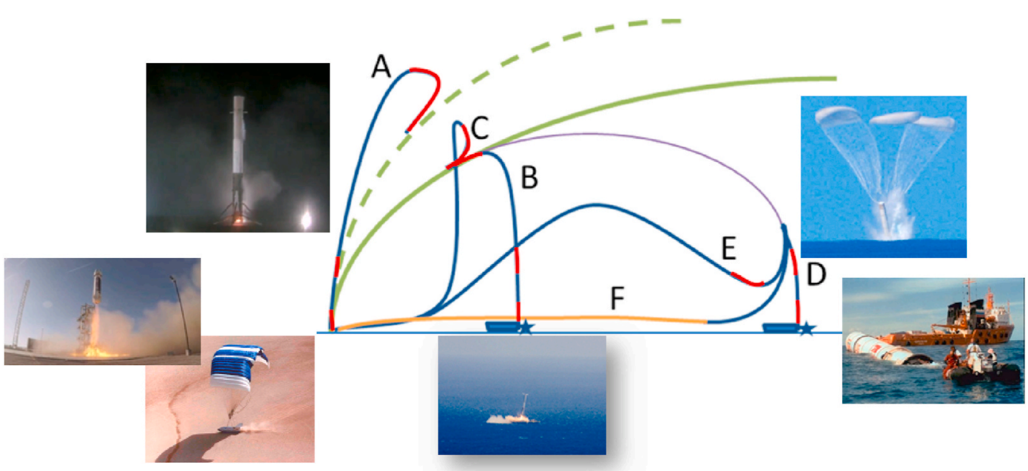


Fig. 2. Possible recovery scenarios [10].

1961 and a reduced scale flight test was performed in 1963 by NASA Langley research center (Fig. 3). In this frame a reduced scale test of paraglider recovery system of the Saturn IB booster stage (WTT in 1961, flight test in 1963 by NASA Langley research center) was performed and then abandoned. Finally, the paraglider has been used for other applications (example of space technology spin-off). Another example of the early space age is the recovery of the Gemini V liquid booster for engineering expertise in 1965 (Fig. 3).

However, the first operational booster stage recovered has been the space shuttle Solid Rocket Boosters (SRB) (Fig. 4). The initial space shuttle design foresaw a completely reusable system composed of a booster and an orbiter stage in piggy back. Different reasons took to the decision of implementing SRB, a central tank and a cross feeding to the space shuttle main engine (SSME). This new design permitted to "reuse" all the launcher parts except the central tank. The SRBs were recovered after splashdown and then refurbished [14], but the latter phase revealed to be quite costly (22 M$ in FY-1986 following [5] to refurbish two SRBs). Two ships were used to retrieve the two SRBs after splashdown.

Ariane 5 Solid boosters (EAP) were also designed to be recovered after ballistic phase and then splashdown, but only for expertise purposes and on specific flights. The recovery system was based on a classical chute chain with a final subsonic descent using three main chutes (Figs. 4 and 5).

During space age, a number of different concepts has been developed to reuse parts of the launch vehicle. For instance, Dornier investigated the recovery of Ariane 5 liquid propulsion bay in 1990 (project RRL), ULA made investigations to recover and reuse) the main propulsion engines by means of inflatable heat shield (HIAD Hypersonic Inflatable Aerodynamic Decelerator), guided parafoil and helicopter capturing [15]. A large scale, high energy flight demonstration concept for HIAD has been developed and tested for atmospheric re-entry.

Airbus-DS company proposed the concept Adeline (ADvanced Expendable Launcher with INnovative engine Economy, 2015), whose goal was to recover the main propulsion and avionics using fly-back. The propulsion bay was part of a winged aircraft deployed after MECO and able to perform a maneuver and the a trimmed flight to reach the launch site [16]. Reduced scale tests of the Adeline vehicle and its sub-systems were performed during preliminary design phases.

In past years the Russian Federation proposed a number of liquid reusable stages for the launch vehicle Angara, mainly in order to avoid fallout of the stages on ground. The Baikal reusable booster stage (2001) was based on Liquid Oxygen (LOX)/kerosene propellant couple and powered by the engine RD-180, RD-191 depending on the sizes and applications [8]. The peculiarity of this vehicle was the capability of the main wing to be folded at take-off and deployed for cruising flight at subsonic speed. The boosters' separation occurred at about Mach 7. On the base of the flight envelope, Baikal has been characterized by a complete aerodynamic test campaign in Russia to build a data base including thermal maps of the vehicle (Fig. 6). The final design of the Angara stages adopted the solutions of fully expendable components. The same principles have been used for the joint CNES/TSNIIMASH project *Bargouzin*, implementing two reusable fly-back liquid boosters to Ariane 5 [9].

The concept proposed by Kistler Aerospace (2001) was a fully reusable TSTO, employing unwinged stages and low separation Mach (about 4) to easily retrieve the LOX/kerosene booster stage (Fig. 7). It was a small launcher class, delivering about 2600 kg to 52 degree-800 km orbit. Each stage carried its own suite of redundant avionics and was supposed to be operated autonomously. Nominal stage separation was designed to occur at about 40 km of altitude at a velocity of 1200 m/s. After stage separation the booster stage had to be restarted to fly back to launch site and had to be retrieved by parachute chain and safely recovered by means of deployed airbags. The orbiter was supposed to deliver the payload and perform a synchronization to re-enter and be recovered on the launch site. Thus both stages were supposed to be recovered on the launch site. This project never went up to flight, but some principles are very similar to the approach used by SpaceX to retrieve the Falcon 9 boosters.

Recent years have been marked by the proven boosters' toss-back/barge-landing reusability of Falcon 9 and New Shepard [18]. Even if the two systems are completely different in terms of mission, goals and energies, they use a similar approach for recovery based on the engine re-ignition and throttling capabilities. One of the key features of the toss-back consists in avoiding to add wings to the stage structure, which was the peculiarity of most of the reusable concepts to date (Fig. 8). This booster stage recovery strategy is based on previously existing ideas, but it has been taken to operational flight, including the multiple reuse of booster stages and even of the fairing. For instance, main propulsion maneuvering to glide back to the launch base was the investigated by Aerospatiale company under CNES contract back in 1984 in the project called FLS [17]. In this case the booster stage was equipped with small delta wings and rudders.

Other national programs decided to investigate the interest of a similar approach for future launch systems. This is the case of China which investigates a completely reusable first stage, with powered descent for the core using leftover fuel to land vertically and parachutes for the boosters on the Long March 8 launch vehicle [20]. In August 2020 the Chinese Company LinkSpace performed a successful hop test of its suborbital reusable rocket (SRV-1) powered by a LOX/LCH4 (Methane) engine [21].

This is also the case of France, Germany and Japan through the recent experimental project Callisto, whose main goal is to prove critical technologies to be matured for flight [22] (more information on the main objectives of this demonstration are provided in Section V). Callisto is powered by a 40 kN LOX/LH2 (Liquid Hydrogen) expander

Fig. 3. SATURN IB paraglider WTT in 1961, Gemini V booster in 1965.

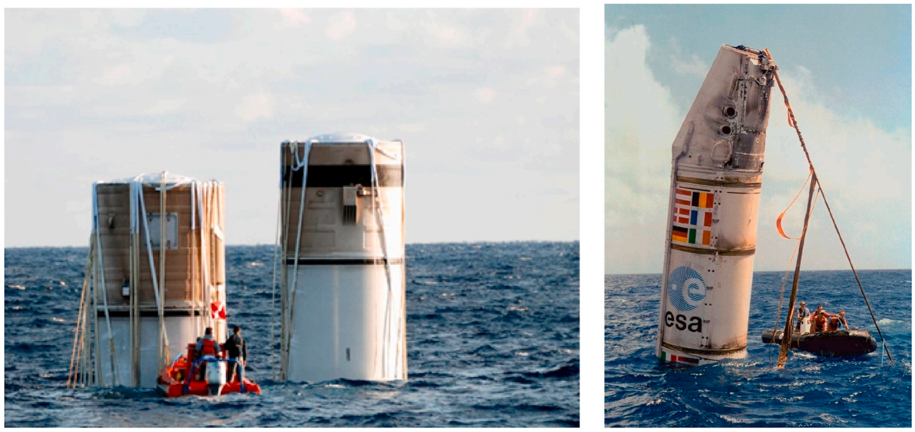


Fig. 4. Space shuttle SRB and Ariane 5 EAP splashdown.

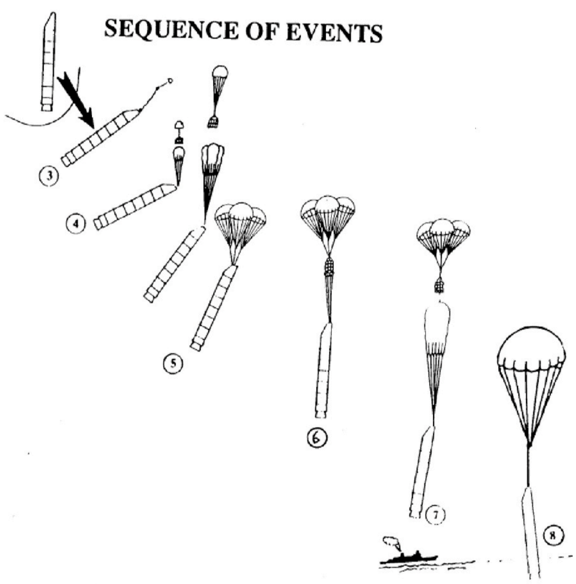


Fig. 5. Ariane 5 EAP recovery sequence after separation (Dutch Space courtesy).

cycle reignitable engine developed by ISAS/JAXA [23,24] and its operational phase includes 10 flights over a period of 6 months (Fig. 9) [25]. Themis is the demonstrator stage developed by the French Airbus/CNES company ArianeWorks powered by a set of low cost LOX/LCH4 1000 kN reusable and throttable Prometheus engines. CNES has already flown the FROG system to test some features of Guidance Navigation and Control (GNC) through an ad hoc hop test (2020) (Fig. 9).

With respect to the scheme shown at the beginning of this Section III, solutions used by SpaceX are A and B, while solution D has been employed for SRB and EAP (and occasionally by SpaceX). Recovery after splashdown imply effects of the sea water on the stage, as well as wave impact, and may require a difficult refurbishment phase, i.e. costly and time consuming. The "soft" recovery of solutions A and B present the disadvantage to require a precise recovery, but they have an easier refurbishment phase. However, the technology to be developed to reach that goal require important investments and test campaigns to secure the flight. In addition, depending on the mission, the trajectory must be deflected to allow the toss-back of the booster stage (trajectory "verticalization") and this translates into payload loss, which can be applicable to low filling ratio missions. Recovery with paraglider is only possible for small mass stages, and in practice impossible for heavy class launch vehicles.

New Sheppard by Blue Origin recovers a space tourism low energy stage with precise LOX/LH2 engine powered landing. The stage follows a typical ballistic trajectory after delivery of the manned capsule.

3. High energy spacecraft re-entry

In the frame of the high energy reusability, i.e. the recovery of orbited spacecraft, the vehicles can be divided into two main categories [26]:

- Winged/lifting bodies, capable to precisely target a landing runway or a landing site.
- Apollo/Soyuz like capsules using parachute systems to perform landing or splashdown.

3.1. Winged/lifting body

Concerning the first kind of spacecraft, the space shuttle program took advantage of numerous experimental vehicles, such as ASSET, X-15, X-23, X-24. An important experience in the hypersonic flight domain had been gained with X-15 (about 200 flights) in terms of thermal metallic protection, TAEM (Terminal Area Energy Management) management, non-propelled landing for a vehicle with a poor Lift/D ratio. With ASSET and PRIME orbital and sub-orbital re-entry flights, FCS (Flight Control System) and RCS (Reaction Control System) efficiency, thermal protection system (metallic), aerothermodynamic measurements, flight worthiness, guidance accuracy have been explored. With X-24 (USAF), a landing training vehicle with poor aerodynamic characteristics, transonic controllability has been investigated. X-38 had a shape very close to the X-23 and was supposed to be a scaled model of the Crew Rescue Vehicle to the international space station. ESA has been associated to this NASA program, which actually stopped in 2002. The X-38 allowed Europe to finalize the body shape and gain experience in the frame of nose and body flap technology as well as in the aerodynamic characterization at high speeds, shape design, GNC and Thermal Protection System (TPS) architecture (Fig. 10).

Russian in-flight experimentation was mainly based on two experimental vehicles developed in the frame of the BURAN program. Those are BOR-4 (flown between 1982 and 1984 up to Mach 25) and BOR-5 (flown between 1983 and 1988, up to Mach 18). The Russian strategy consisted in splitting the reentry demonstration in two missions (consequently two vehicles) to avoid the management of the compromise between aerothermodynamic problems and aerodynamic ones, due to the scale effect of the demonstrator. As a result of that logic, BOR-4 was dedicated to orbital re-entry with a shape representative at scale

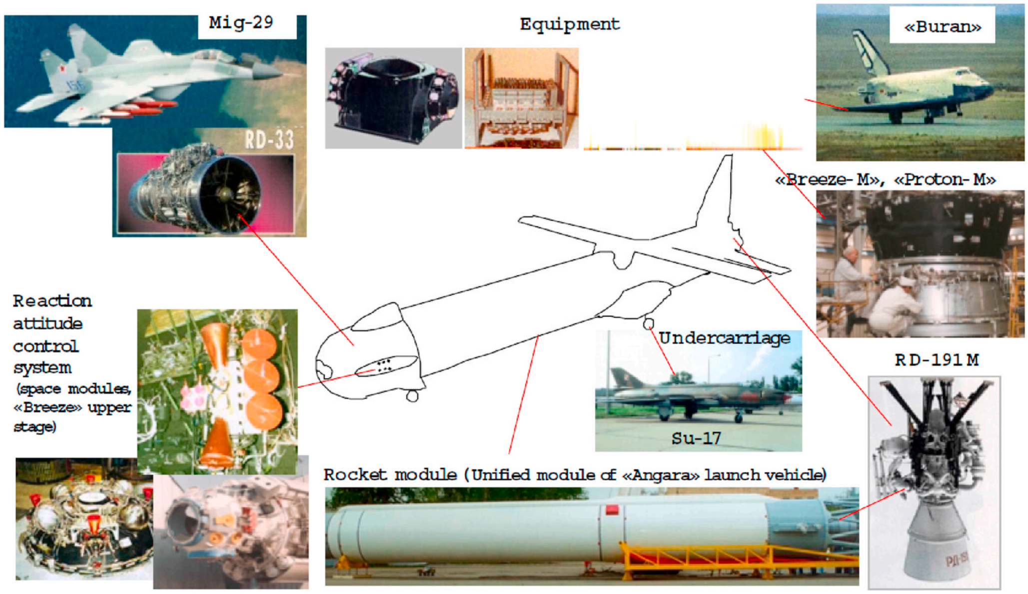


Fig. 6. Main components of Baikal (Khrunichev-Molniya courtesy).

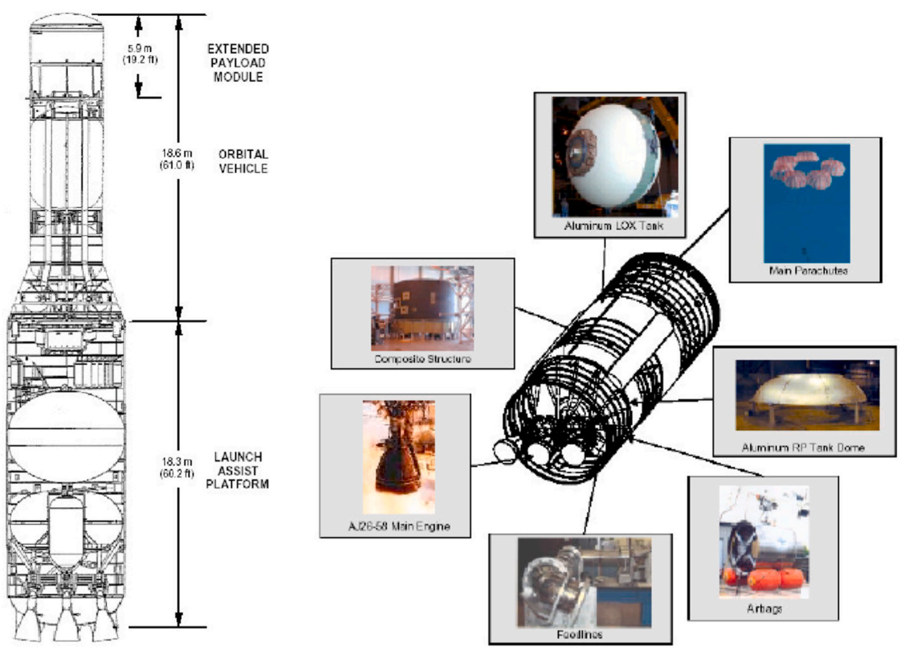


Fig. 7. Kistler K1 LV and LAP booster (Kistler Aerospace).

1 of the BURAN nose/windward curvature. The BOR-5 was a subscale of BURAN dedicated to aerodynamic efficiency, flight worthiness and GNC (Fig. 11).

In Japan, an important fleet of experimental vehicles has been developed to support the development of the HOPE-X program. The HYFLEX (Fig. 11) and HSFD have been developed in this frame, but the high hypersonic domain controlled flight has not been explored.

Finally, Fig. 12 summarizes the flight domain and the number of flight for each USA X vehicle. It has to be noticed that there was no space shuttle sub-scale demonstrator in the development program. Only the first prototype 'Enterprise' was used for in-flight demonstration. Those experimentations were limited to TAEM and landing mastering after release from a modified Boeing 747.

A large part of these experimentation programs prepared the technology needed for space shuttle (Fig. 13), whose flights provided a huge amount of data recorded in Ref. [27] for what concerns the aerodynamics, aerothermics, TPS management, in-flight measurements. Its design, based on the combination of a booster and orbiter stage, required the mastering of several technologies: the large solid rocket boosters SRB, the cross-feeding to pass the fuel and oxidizers to the space shuttle main engine (SSME), the staged combustion reusable engines, the management of reusable protection system (TPS), the re-entry automated guidance and control (GNC) and, not last, the management turnaround operation of a fleet of space vehicles. The space shuttle required extensive inspection and refurbishment. In particular, the thermal protection tiles and gap fillers [28] were individually inspected (and eventually replaced), and the SSME main engines were removed for extensive inspection and overhaul [29]. The SRB were contaminated with ocean salty water and had to be cleaned, disassembled, and refurbished before reuse [14].

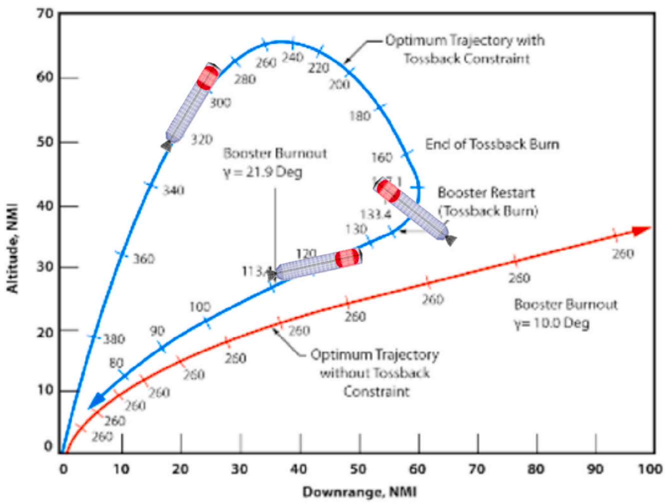


Fig. 8. Typical booster stage toss-back mission [19].

On the other side, Soviet Union developed the Energia-Buran system, based on the same shape of the shuttle, but using a liquid main propulsion and boosting, both expendable.

With its 135 missions, space shuttle produced a large amount of information, results and experiences that paved the way to future reusable vehicle programs. However, even after space shuttle, in-flight experimentation remained an essential step to mature new technology. For instance, between 1990 and 1993, the Delta Clipper Program led to develop a sub-scale experimental vehicle, the DC-X, improved later by NASA with the DC-XA, that performed 8 successful flights before destruction. This program allowed to gain experience on Thermal Protection System (TPS) technology, LH2 composite cryogenic tank and Al–Li LOX tank technology and on operations with reuse of the same vehicle with only light ground infrastructure and a turnaround capability of less than 48 h.

This long heritage finally allowed the realization of the operational X–37B, accomplishing an in-orbit mission of more than 400 days (Fig. 14). The operational vehicle has been possible thanks to the experimental vehicle X-37, a test bed that boarded more than 40 innovative technologies. A LEO service is proposed by the Dream Chaser vehicle [30], whose shape has been directly derived from the Russian experimental lifting body BOR4 (this can be easily seen from comparison of Figs. 11 and 14). BOR-4 had excellent re-entry characteristics, from Mach 20 down to high transonic speeds and needed little control surface deflection to remain stable [26].

In Europe the HERMES program (Fig. 15) has been the first concrete development to conceive a reusable vehicle with a Shuttle/Buran like technology. Even though this program was cancelled, it permitted the development of knowhow, technologies, computer codes and testbeds. Typically, the high enthalpy wind tunnels, the reusable TPS, the algorithms for guided re-entry are some example of the HERMES heritage. After HERMES the civil re-entry studies had been continued by other ESA, ESA/NASA and national programs such as FESTIP (ESA), X-38 (NASA with ESA participation), FLTP (ESA) and ARD (ESA). These programs permitted to continue developing the key technologies for re-

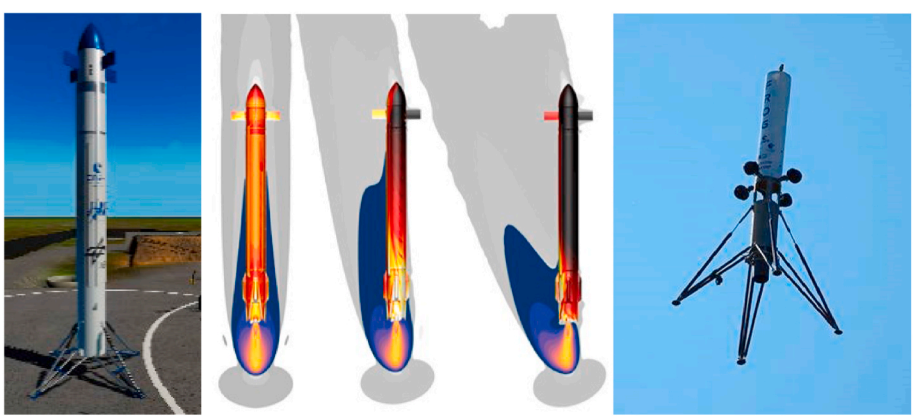


Fig. 9. Callisto configuration and CFD re-entry motor on [24] (CNES/JAXA/DLR) and the recently tested FROG (CNES).

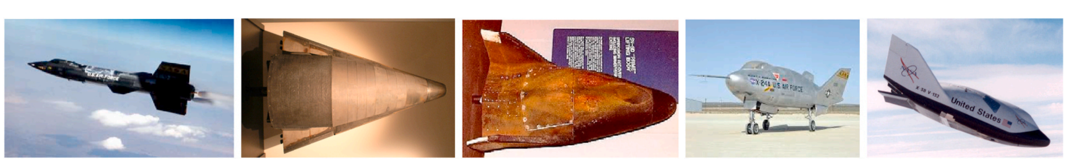


Fig. 10. X-15, ASSET, X-23 (PRIME), X24 and X-38.

Fig. 11. BOR-4, BOR-5 and HYFLEX

Flight domain	Vehicles	Number of flights	Total
M=7 to M=25	ASSET	6	9
	PRIME	3	
M=2 to M=7	X-15	199	199
M=1 to M=2	X-24A	9	80
	X-24B	21	
	M2-F3	26	
	HL-10	24	
M<1	X-24A	19	146
	X-24B	15	
	M2-F1	77	
	M2-F2	16	
	M2-F3	1	
	HL-10	13	
	Enterprise	5	

Fig. 12. In-flight experimentation status in 1980.

entry.

ESA recently launched the Space Rider program, based on the IXV experimental re-entry lifting body heritage, in order to perform LEO operations. IXV allowed to test a number of critical technologies for re-entry lifting bodies including reusable thermal protections on nose, body flaps and leeward, re-entry trajectory strategy and precision landing. It flew in 2015 on the VEGA launch vehicle (LV) (Fig. 15). The preliminary design of the vehicle was performed by Airbus under CNES contract (Pre-X) [31] and IXV by TAS under ESA contract. A number of technologies coming from the Hermes program, the X-38 and the test benches made available in Europe, have been used in the frame of Pre-X/IXV. The heritage of this program is huge for the European space community in terms of re-entry technology developments, in particular concerning aerodynamics/ATD (Aero-Thermo-Dynamics) design, TPS technology, GNC and more in general the complete system testing during a full re-entry trajectory in one shot [32]. This allowed Europe to mature re-entry technology up to flight testing.

India has also flown the RLV-TD experimental vehicle, which successfully completed its first atmospheric test flight in 2016 and lasted 770 s reaching an apogee of 65 km. During this mission, the demonstrator was heavily instrumented and excellent flight data were obtained on-board through suitable sensors and flight telemetry system, which gave valuable and crucial details of hypersonic aero-thermodynamics, autonomous navigation, guidance, control, integrated flight management [33].

SpaceX is currently developing the Starship vehicle for Mars mission and its corresponding prototype. The Starship will launch atop the company's Super Heavy rocket and is made of stainless-steel alloy derived from 304. The vehicle will be powered by six Raptors, compared to this prototype's three engines (three will be for sea level and the rest for the vacuum of space). The ground and flight tests of the prototype are currently ongoing [34].

3.2. Capsules

The Soyuz program is the longest operational human spacecraft program in the history of space exploration. The first crewed flight into space was on April 23, 1967. Although they were conceived by the Soviet Union at the start of the sixties, the Soyuz spacecraft are still used today, but with important modifications. They have transported Russian crews to the Soviet stations Salyut and Mir. Since the last shuttle flight, the Soyuz have been the only spacecraft available to transport crews to and from the International Space Station up to 2020. The capsule itself has not been conceived to be reused, but the shell and some components can be re-utilized (Fig. 16). However, even if investigation for this capsule reusability have been considered [35], the system is expendable.

Europe developed the Atmospheric Re-entry Demonstrator (ARD) re-entry experiment based on the same shape as the Apollo capsule, which successfully flew and was recovered after the 3rd qualification flight of Ariane 5 in 1998 (Fig. 17). ARD flight has been a full success and allowed to demonstrate the mastering of aerodynamics/aerothermics, GNC and flight control, TPS, communications and plasma, parachute chain, but the major result was the testing of the overall system in real flight environment. Lessons learned allowed to prepare the future flight demonstrators such as Pre-X/IXV, in particular for what concerns reusable TPS (even if on ARD only samples were tested around a fully ablative patch), GNC and flight control algorithms, aerodynamics/ATD

Fig. 14. X–37B and Dream chaser.

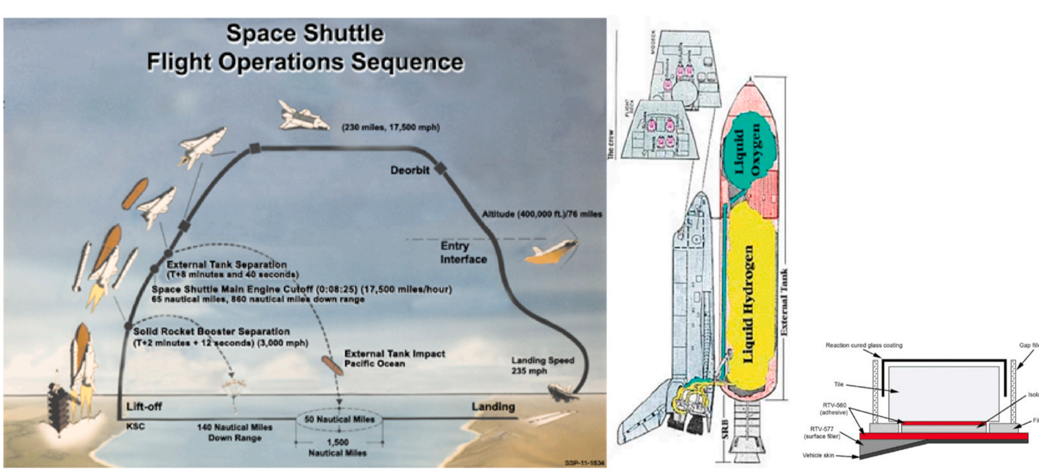


Fig. 13. Space shuttle mission scheme, main propulsion propellant management and tiles (NASA courtesy).

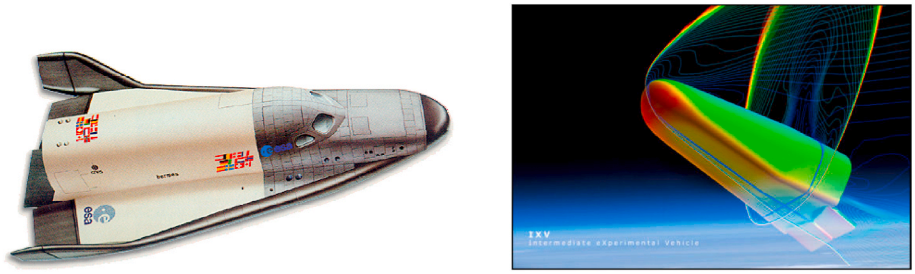


Fig. 15. HERMES spaceplane (left – ESA courtesy) and IXV re-entry lifting body (right - ESA/TAS courtesy) [32].

Fig. 16. Soyuz capsule (Roscosmos courtesy).

tools and test benches [36].

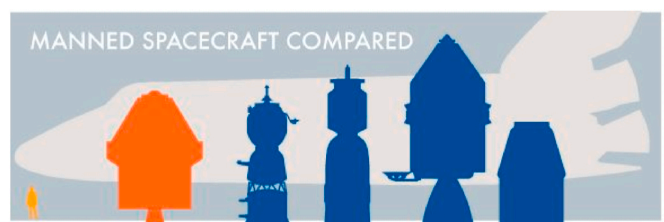
New generation re-entry capsules have been developed to allow manned flight to ISS and to the Moon. The SpaceX Dragon performed its maiden flight to the ISS on May 30th' 2020. For a very long time capsules, both manned or cargo, have been considered as one shot vehicles, but the new generation consider the reusability with suitable refurbishment and verification tests before flight.

The incredible heritage of Apollo capsules and of the ESA Automated Transfer Vehicle (ATV) allowed the development of Orion (officially Orion Multi-Purpose Crew Vehicle or Orion MPCV) [37], a class of partially reusable space capsules to be used in NASA's human spaceflight programs. The Orion capsule consists of three basic sections: a crew module, a service module, and a launch abort system. Basically, it is most of the Crew module that can be refurbished and reused, but not the heat shield [38]. Boeing designed and started testing the Starliner capsule, a class of reusable crew capsules expected to transport crew to the International Space Station (ISS) and to private space stations. The capsule is slightly larger than the Apollo command module (d = 4.56 m) and smaller than the Orion capsule. Starliner is able to remain in-orbit for up to seven months with reusability of up to ten missions and is designed to be compatible with four launch vehicles: Atlas V, Delta IV, Falcon 9, and Vulcan. It succeeded its first airbag landing in December 2019 and uses a lightweight ablator for its heat shield.

Finally some manned spacecraft with capsules are compared in Fig. 18 where the capsules that can be refurbished are essentially Orion and Dragon.

Finally, the re-entry profiles for some vehicles are provided in Fig. 19, where the different aerodynamic/ATD regimes are highlighted in the geodetic altitude versus relative velocity frame. It can be observed that the maximum heating region of the Pre-X lifting body is very close to that of the spaceplane typical trajectory. Fig. 20 shows four vehicles in the same plane: space shuttle, Pre-X, Hyflex and ARD. It is apparent that the ARD re-entry profile is much steeper and Hyflex is suborbital and then has a similar regime to ARD, while Pre-X is orbital with a path following closely that of the shuttle.

The recovery of upper stages has not been developed yet for the launch systems up to the heavy class (excluding huge). Having these stages high energy and relatively low cost with respect to the booster stage, the gain due to recovery can be low, but the cost to mass ratio is



	Orion	Soyuz	Shenzhou	Apollo	Dragon	Shuttle Orbiter
Builder	Lockheed Martin/ESA	RSC Energia	China	Rockwell Int'l.	SpaceX	Rockwell Int'l.
First crewed flight	2021	1967	2003	1968	2014	1981
Crew	2 to 6	3	3	3	7	7
Launch vehicle	SLS	Soyuz FG	Changzheng 2F	Saturn	Falcon 9	STS
Length overall	26 ft (8 m)	24.5 ft (7.5 m)	30 ft (9.25 m)	36 ft (11 m)	20.4 ft (6.2 m)	122 ft (37.2 m)
Liftoff weight	46,848 lbs. (21,250 kg)	15,763 lbs. (7,150 kg)	16,765 lbs. (7,600 kg)	66,871 lbs. (30,332 kg)	26,455 lbs. (12,000 kg)	242,500 lbs. (110,000 kg)

SLS = Space Launch System STS = Space Transportation System

Fig. 18. Manned spacecraft with capsules compared [39].

Fig. 17. ARD re-entry capsule and CMC sample on the heat shield (Airbus courtesy) [36].

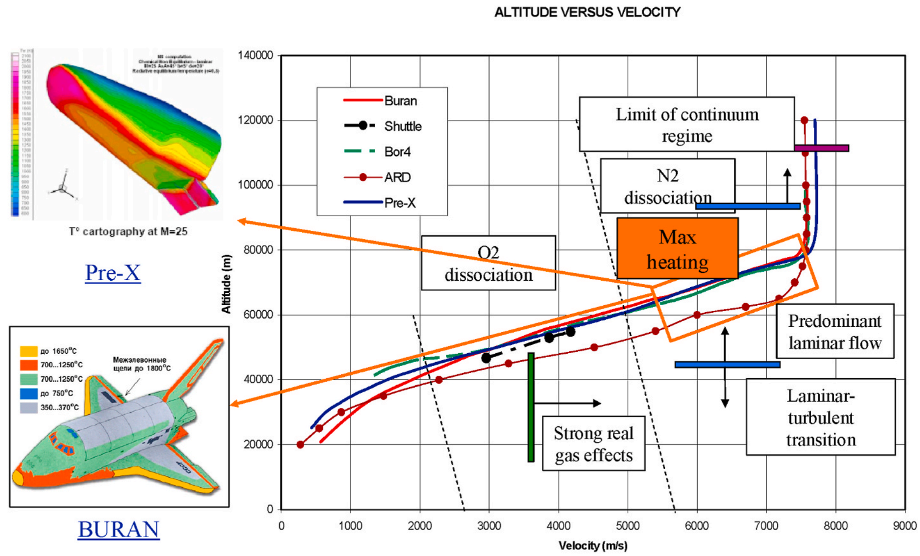


Fig. 19. Typical re-entry trajectory of winged/lifting body vehicles with associated aero-thermal phenomena.

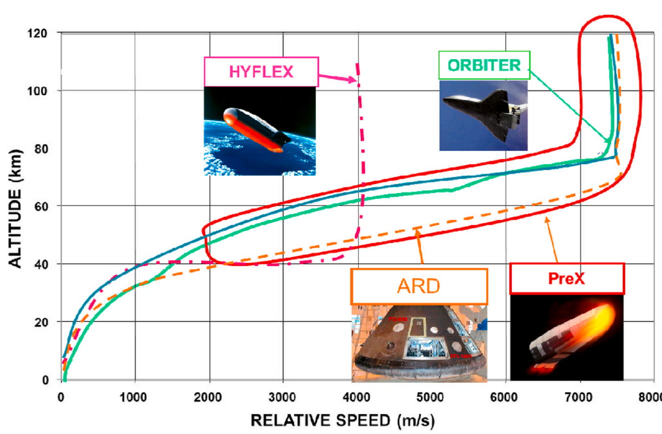


Fig. 20. Comparative re-entry profiles for different vehicles.

usually higher than the one of booster stage. In addition, recovery is easier for LEO missions than for others (e.g. MEO and GTO). The subsystems required for recovery are essentially the de-orbiting system, the heat shield, the deceleration system and the landing/splashdown system. Different choices exist for each one of these subsystems. The heat shield can be either deployable/inflatable or solid. The former option has the advantage to provide aerodynamic stabilization during re-entry through a conical shape as shown for example in Ref. [40]. This kind of system flew few times aboard a Russian Volna launcher (from 2000 to 2005). The IRDT (Inflatable Re-entry and Descent Technology) demonstrator was the result of a co-operation among ESA, DASA, Lavochkin and the European Commission [41] (Fig. 21). This technology enables re-entry from orbit and landing without the need of a heavy heat shield or parachute system and consists of a small ablative nose from which a cone-shaped flexible heat shield inflates to provide thermal protection and braking during re-entry. The flexible heat shield is covered by an inner insulating layer protected by a silica-based fabric impregnated with an ablative material in contact with the hot plasma flow. A second inflatable extension of the cone, deployed during subsonic flight, reduces the speed of the Demonstrator to ensure a safe landing. Possible applications are return of experiments from the International Space Station, Mars probes and return of launchers' upper stages. Even if the flights were non-nominal, this experimental vehicle provided a number of results on the ability of the shield to withstand re-entry thermo-mechanical environment and on the deployment system.

Other conical recovery systems have been developed both for low and high energy trajectories. One example, already discussed in this paper for the recovery of the Atlas Vulcan engines, is given by the HIAD (Hypersonic Inflatable Aerodynamic Decelerators). The HIAD underwent Inflatable Re-entry Vehicle Experiment (IRVE) tests on a sounding rocket. When mature, this technology could be used for planetary entry. The advantage of inflatable re-entry structures resides in the compact volume during storage on the launch vehicle and reduced mass during

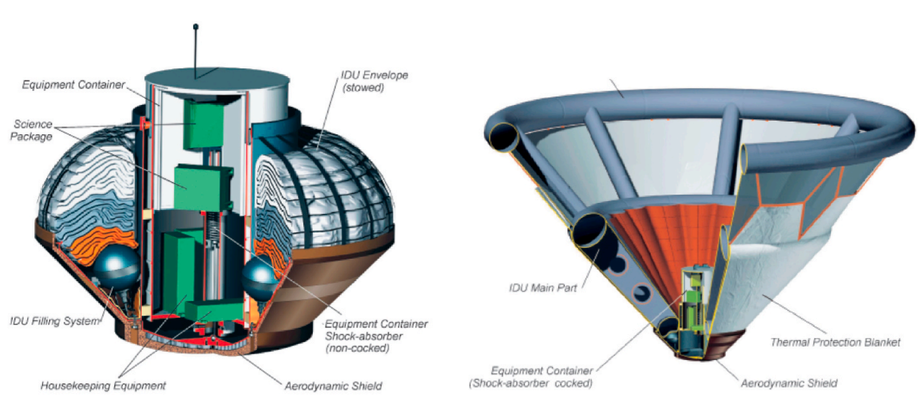


Fig. 21. Inflatable Re-entry and Descent Technology (IRDT) demonstrator [41].

re-entry [42].

4. Technology aspects

This Section focuses on technology aspects only based on European programs (except for the propulsion part) and it is split in the following paragraphs: critical trajectory phases, system technology and demonstration, some subsystem technology and propulsion.

4.1. Critical trajectory phases (orbit or ballistic)

In order to better identify the re-entry features of the different kind of vehicles, typical flight profiles in the altitude-velocity plane are given in Fig. 22 for booster reusable stages as well as for orbiters re-entry vehicles (the values are just indicative). A corresponding profile in the plane acceleration-velocity can be obtained giving the constraints represented by the load factor, the thermal loads, the equilibrium gliding and the maximum dynamic pressure. The relative velocities and Mach numbers are represented on the graph as well as the maximum heat fluxes, the maximum dynamic pressures profiles, the Reynolds number and continuum regime curves. This representation of the re-entry trajectory allows to identify the different physical domains faced during descent and the corresponding critical areas in terms of physical phenomena, such as heating and dissociation for high energy vehicles. It is also possible to demonstrate in-flight or on-ground the critical phases in order to secure full scale flight. The re-entry trajectories are traced for Hermes, Buran and a typical booster and orbiter of a two stage launcher.

4.2. System technology and demonstration

The flight domain of a typical booster stage vehicle is represented in Fig. 22 for a typical fly-back winged vehicle. The thermo-mechanical loads are directly function of the separation Mach (and then re-entry Mach), the vehicle shape and the angle of attack (AoA). Critical areas for a typical booster stage re-entry are the following:

- Maximum heat flux, occurring at altitudes and re-entry Mach function of the separation Mach (typical altitudes are 30–40 km).
- Maximum acceleration and maximum dynamic pressure. This is different if the re-entry is ballistic, boosted or winged with turning maneuver.
- Recovery area, depending on the retrieving system, the final braking approach and Guidance Navigation and Control (GNC) to allow automatic landing/splashdown.

In the case of orbital re-entry vehicle, the critical areas are occurring at maximum heat flux, whose corridor depends on the thermal protections and associated chemical phenomena. The corresponding reduced scale experimental vehicles can cover typical critical areas of the trajectory. European wind tunnels allow to test typical re-entry conditions in terms of Mach and enthalpy in part of this trajectory range, usually below Mach 10 (e.g. ONERA R2Ch, DLR HEG, ONERA F4, CIRA Scirocco and Ghibli).

For an experimental vehicle with a characteristic length L (e.g. the nose radius of the vehicle) and crossing the atmosphere with the density ρ, the driving parameters of the aerodynamics and ATD phenomena are:

- The Mach number M, which gives the flow compressibility.
- The Reynolds number Re, which gives the ratio between the inertial and viscous forces.
- The dissociation parameter ρL, representing the gas dissociation and hence to real gas effects.
- The heat flux per unit area at nose stagnation point Φ.

The similarity parameters of Fig. 23 must be evaluated with respect to the geometric variables such as $\lambda = L_{x\text{-vehicle}}/L_{ref\text{-vehicle}}$, where $L_{x\text{-vehicle}}$ is a characteristic length of the experimental vehicle and $L_{ref\text{-vehicle}}$ is the correspondent characteristic length of the scale 1 vehicle. Together with the geometrical factors, a trajectory factor can be defined as the ratio of the density of a reference scale 1 vehicle and that of the scaled model:

$$K = \rho_{ref\text{-}vehicle}/\rho_{x\text{-}vehicle}$$

Supposing that the scale 1 and experimental vehicle have the same velocity, similarities are respected when $\lambda/K = 1$ for aerodynamics and $(\lambda \cdot K)^{0.5} = 1$ for thermal flux. Fig. 24 provides the similarity conditions for aerodynamics and heat flux for flight of the experimental vehicle at the same velocity, but at different altitudes with respect to a reference

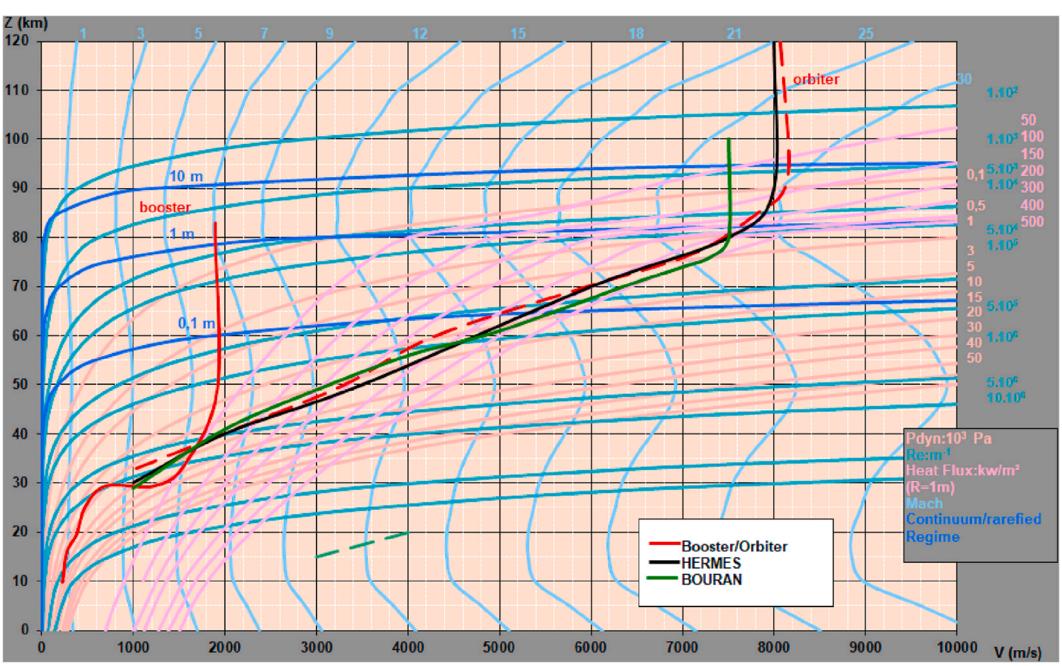


Fig. 22. Typical re-entry profile represented on altitude Z – velocity V plane.

Similarity Parameter	Characteristic Dimension L	Physical Phenomena	Relevant Similarity
Reynolds Number $\mathrm{Re} = \frac{\rho.V.L}{\mu}$	Body Length	Viscous Effects on Flaps (SWBLI)	Major
	Nose radius	Viscous Effects on Nose / Forebody (Transition on windward side)	Major
Dissociation Parameter (Binary Mixture) $\rho.L$	Body Length	Real Gas Effects (Body Flap)	Minor
	Nose radius	Real Gas Effects behind Front Shock (Stagnation Region) - Catalycity	Major
Heat Flux $\Phi_{Nose} = C.\sqrt{\rho}.\frac{V^3}{\sqrt{R_{Nose}}}$	Body Length		Not relevant
	Nose radius	Nose Heat Flux	Major

Fig. 23. Aerodynamic and ATD similarity parameters [26].

vehicle, as a function of λ. A difficulty of defining the shape of a scaled vehicle is to determine with respect to which reference vehicle this has to be defined.

European technology and test means for high energy reusable vehicles are often issued from the heritage of the HERMES spaceplane. In particular, Fig. 25 shows the main items developed in the frame of this program and then used for the ARD, Pre-X and IXV re-entry vehicles.

In the case of boosters stages, problems linked to mechanical energy, flight path angle (FPA), and thermal constraints are less important with respect to those of an upper stage or orbiter. The level of complexity for retrieving a booster stage depends mainly on the couple altitude-velocity vector reached at stage separation. For a two stage launcher, in the case of LEO missions the separation altitude may vary between approximately 70 and 85 km, the velocity modulus varies between approximately 1500 and 2000 m/s and the FPA between 30 and 70°. The corresponding separation Mach number varies between 5.5 and 6.5. For GTO trajectories the velocity modulus increases up to 2300 m/s at about 70 km with a FPA = 30°, with a corresponding separation Mach number of about 7.5. Of course these are typical values which depend on launch vehicle staging and mission strategy. The separation Mach number, altitude and downrange are driving factors for "toss-back" or "barge landing" missions.

Critical technologies are strongly dependent on the energy reached by the vehicle and can be split between the rocket boosters and re-entry spacecraft (or stages). The main technologies concerning re-entry is oriented towards orbital re-entry, such as reusable TPS, aero-thermo-dynamics phenomena, GNC and refurbishment [44]. Other technologies are applicable to reusable boosters, such as reusable propulsion, Health Monitoring System (HMS), GNC, recovery and refurbishment.

The refurbishment phase is key for any reusable vehicle since it is directly linked to the cost and to the turn-around. Its associated effort is strictly linked to the HMS, particularly for what concerns the main propulsion and structures. In case of space shuttle this effort was important and essentially devoted to the TPS and to the main propulsion (SSME). In the case of SRB, this phase consisted of a "cleaning" phase (because of splashdown) followed by removal of degraded expendable parts and almost complete dismounting of the solid booster, before refurbishment [14].

Parachute retrieval is indeed an interesting alternative to be considered since potentially leading to less complex and then less costly

For X or RLV Similarity : Pré-X should fly		
In Case	Aerodynamics	Heat Flux
λ < 1	Lower Z	Higher Z
λ > 1	Higher Z	Lower Z

Fig. 24. Similarity conditions versus λ, Z [43].

launch systems. In this case the precision landing can be obtained only by means of guided descent system and wind prediction before flight. In any case it cannot be as precise as the one obtained by controlled paragliders, fly-back or boost-back systems. However, controlled paraglider is currently possible only for small mass spacecraft.

In case of fly back maneuver of the booster (winged) stage to return to launch site, aerodynamic perturbations, propellant feeding during deceleration and fuel sloshing are main issues to be handled during this maneuver. A dedicated analysis of the fly-back maneuver and associated sequence of operations is needed. The main critical technologies of toss-back boosters are the specific engine design (reusable, throttable, re-ignitable …), the precision GNC, the propellant management as well as the landing synchronization in case of multiple boosters. Some of these technologies can be secured by on-ground and in-flight experiments.

Concerning the demonstrators and experimental vehicles, Callisto is given as example in the frame of CNES activity in preparation of future launch systems. The Callisto demonstrator objective is to obtain flight and ground data to verify the feasibility and interest of a future reusable booster stage [24]. The overall goal is to demonstrate the ability to recover a vehicle in toss-back and vertical-landing modes as well as turnaround operations linked to refurbishment and reuse with the aim to explore the overall mission cycle of a reusable first stage. On CNES side there is the additional goal to demonstrate the capability to operate a reusable vehicle, representative of a booster stage at the European spaceport of Kourou. The general features of the demonstration are the following:

- Test a complete system in-flight.
- Experimental vehicle philosophy, i.e. a flying test-bed with technical and technological objectives, but not fully representative of an actual vehicle.
- Main goal: explore part of typical booster stage flight envelope, test critical areas/technology and verify recovery & turnaround cost hypotheses.

These goals have been translated into requirements for the system and a constraint, the use of the LOX/LH2 reusable engine provided by ISAS/JAXA.

For toss-back booster stages requirements on propulsion are the most demanding and are shown in the dedicated paragraph of this Section. However, precision GNC, maneuvering required after Main Engine Cut-off (MECO) to toss-back, aerodynamics and propellant mastering constitute the essential of critical technology. Most of the in-flight demonstration activity is dedicated to these aspects. The same holds for barge landing especially if a down range limitation is required [45].

Technology/test beds	HERMES	Pre-X/IXV	ARD
WTT/ high enthalpy tunnels	ONERA R2Ch, DLR HEG, ONERA F4, CIRA Scirocco	ONERA R2Ch, DLR HEG, ONERA F4, VKI, CIRA Scirocco and others	ONERA, DLR
GNC	Precision guidance, navigation, combined aerodynamic/propulsion attitude control	Precision guidance, navigation, combined aerodynamic/propulsion attitude control	Capsule like guidance, navigation, test flight control algorithm during re-entry
Hot thermal protections	Shingles, Carbon-Carbon, Carbon-SiC, FEI	Shingles, Carbon-Carbon, FEI, ablative	Ablative thermal protections and reusable thermal protection samples
Body flaps	Maneuvering, aerodynamics and ATD	Maneuvering, aerodynamics and ATD	-
Blackout management	Radio communications during atmospheric re-entry	Radio communications during atmospheric re-entry	Radio communications during atmospheric re-entry
Final descent, parachute chain	-	Test performances of the parachute and recovery system	Test performances of the parachute and recovery system

Fig. 25. Technology and test means for high energy re-entry.

4.3. Subsystems technology

In case of orbital spacecraft, autonomous GNC is used to manage the re-entry phase and automatic splashdown/landing (on conventional runway for winged/lifting bodies). In Europe this technology for re-entry bodies can be considered as validated by the IXV and ARD flights. A key technology that has been demonstrated in real conditions is the Thermal Protection System (TPS) of the vehicle. ArianeGroup, under ESA contract, has been in charge of the TPS of the windward and nose assemblies of the IXV vehicle, and has developed and manufactured SepcarbInox® Ceramic Matrix Composite (CMC) protection systems that provided a high temperature resistant non ablative Outer Mould Line (OML) for enhanced aerodynamic control [46]. This technology is currently foreseen for the Space Rider vehicle, where the operational and maintenance aspects will be assessed [48]. This TPS is based on the "shingle design", which dissociates thermal and mechanical functions. A thin heat resistant shell made of ceramic matrix composite (CMC) is designed to withstand mechanical loads due to extreme heat fluxes while maintaining the outer aerodynamic line of the vehicle [47]. In addition, layers of insulation material underneath this skin absorb the heat load, and protect the cold structure from high temperatures (Fig. 26).

In case of liquid boosters, reusability implies additional mass to the system, e.g. residual propellants needed for boost-back and associated sub-systems. This induces loss of performance that needs to be compensated by more efficient systems, i.e. better structure ratio for the stages and propulsion I_{sp}.

Propulsion is key for reusability of any rocket stage. The main technological issues, potential difficulties to circumvent in order to design a reusable engine depends on the thrust level, the engine cycle and the propellants' choice. A major choice is the propellant couple. Usually three possibilities are considered: LOX/LH2, LOX/RP1, LOX/LCH4. The LOX/RP1 allows better structure ratios of the stage while having less performing I_{sp} with respect to LOH/LH2, which can have the best I_{sp}, but quite bad structure ratio. In any case the lower the I_{sp}, the higher the structure ratio needed to get the same performance. In toss-back reusable rocket stages, the propellant management to get back (or to land on barge) is essential for staging efficiency. The engine cycle and reusability requirements can improve or degrade the I_{sp} level. In fly-back boosters it is more the structure ratio due to wing-tail structures that matters. The 1st stage efficiency can become quite important for two stages vehicles. In this case it is mandatory to be efficient in both stages propulsion and structure/residuals ratios. An example of structure ratio versus vacuum specific impulse of the booster stage is given in Fig. 27 for given characteristics of the LV upper compound (upper stage plus payload, payload structures and fairing). This results by solving the Tsiolkovski equation with respect to the structure ratio defined as the separated mass (m_s) over the stage overall mass ($m_s + m_p$, being m_p the useable propellant mass).

For a two stage vehicle, the Tsiolkovski equation can be written as follows:

$\Delta V_{id} = -v_{e1} \ln [\varepsilon_1+(1-\varepsilon_1)\pi_1] - v_{e2} \ln [\varepsilon_2+(1-\varepsilon_2)\pi_2]$where

$\varepsilon_k = m_{sk}/(m_{sk} + m_{pk})$

$\pi_k = m_{0k+1}/m_{0k}$

being $k = 1,2$ the number of stages, m_{sk} the kth stage separation mass, the m_{pk} the kth stage useable propellant mass, m_{0k} the LV mass at burnout of kth stage, ΔV_{id} the ideal velocity increment (i.e. the total final velocity provided by the LV) and v_{ek} the ejection velocity of k_{th}

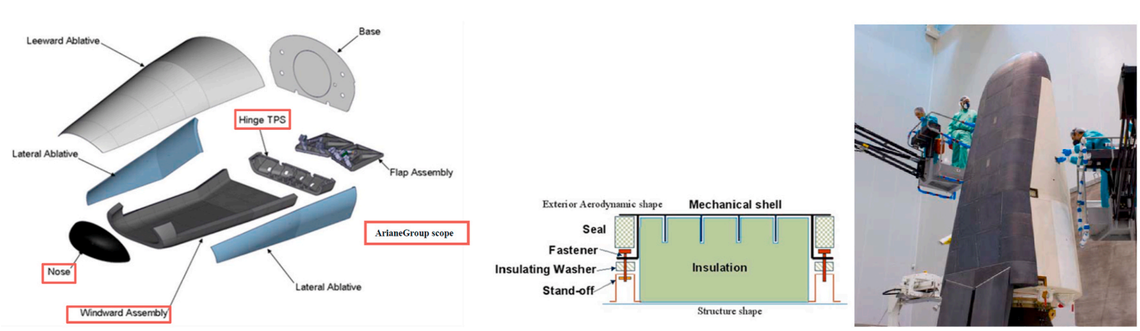


Fig. 26. IXV TPS of windward and nose assemblies (ESA/ArianeGroup courtesy) [47].

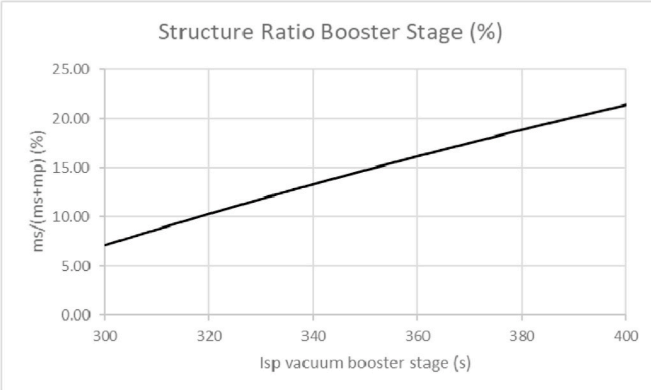


Fig. 27. Theoretical structure ratio versus Isp for booster stage.

stage.

This equation can be solved in order to get ε_1 versus I_{sp1} fixing the other parameters, i.e. the upper compound parameters of the LV. This provides the following closed form equation:

$$\varepsilon_1=(e^{-[c0/(Isp1g0)]}-\pi_1)/(1-\pi_1)$$

where.

$c_0 = \Delta V_{id}-\Delta V_2$

$\Delta V_2 = -v_{e2} \ln [\varepsilon_2+(1-\varepsilon_2)\pi_2]$

$\Delta V_{id} = v_o + \Delta V_{losses}$

being v_o the required orbital velocity and ΔV_{losses} the total velocity increment due to vehicle losses (gravity, aerodynamics, gimballing …).

Considering a LEO orbit with assumptions on payload and upper stage parameters, this provides the function depicted below, providing the structure ratio of the booster stage versus the specific impulse of the same stage. This equation gives the required technology efficiency of the booster stage versus its propulsion efficiency. The higher the efficiency of the upper compound, the lower the technology efficiency of the boosters' stage for provided propulsion efficiency. In the example below 10 s more of I_{sp} allow a degradation of 3 % in structure ratio absolute value.

4.4. Propulsion

In terms of technology needs, propulsion system for reusable launchers needs more powerful and efficient turbo-machinery with increase of reliability and low maintenance requirements. In order to be ready to fulfill these requirements, for both oxidizer and fuel turbo-pumps, it is necessary to develop technologies in the area of inducers, advanced seals, bearings, axial loads balancing systems, impellers, and turbines. In the case of LOX-LCH4 turbo-pumps, the use of fuels having a density comparable to liquid oxygen, introduces the possibility to use a single shaft and direct drive arrangement. This layout can guarantee an engine compact design, overall mass savings, and active balance of the axial thrust by proper re-inverse installation of the LOX and LCH4 impellers.

Health Monitoring is considered an important feature of next generation propulsion systems and for improvement of existing ones. It will be a key element of reusable launchers. The Real-Time context is of outmost importance for the application of HMS on board.

Some of the reusable engines (flown, flying or in development) are listed in Fig. 28, where their main features are outlined.

Aside the well proven and powerful SSME, SpaceX developed the Merlin and Raptor using respectively LOX/Kerosene and LOX/LCH4. New Sheppard preferred the LOX/LH2 propellant couple to power its Sheppard flights (BE-3) and foresees further development for the orbital operational vehicle New Glenn. In this latter case the booster stage is recovered on barge. The BE-4 is a liquefied natural gas (LNG) throttable rocket using an oxygen-rich staged combustion cycle (2400 kN). China private companies developed the 150 kN JD-1 engine, which completed a 200 s hot fire test in 2019. This engine is supposed to power the reusable Hyperbola-2 LOX/LCH4 launcher (1900 kg LEO).

ArianeGroup is developing the Prometheus engine LOX/LCH4 currently under ESA contract and initially under CNES responsibility, a forerunner of about 1000 kN vacuum thrust for the next generation Ariane launchers. This reusable engine is conceived to be low cost and reusable and quite a number of components are manufactured using 3D printing technology [49]. Finally Fig. 28 shows the small reusable JAXA/ISAS engine that shall power the Callisto booster stage experimental vehicle developed in the frame of CNES/DLR/JAXA cooperation.

5. Testing philosophy and objectives

In order to secure developments using new technology, testing is mandatory, both on ground and in-flight, both for sub-systems and for the overall system, like it is the case for a reusable booster or a reusable high-energy capsule/space plane.

Most of propulsion critical technology can be tested on-ground and this holds also for a number of S/S technology such as avionics, structures and propellants lines. But when it comes to system topics strongly coupled with flight events, in-flight demonstration reveals to be a very good way to secure operational flights.

This is the case for the Pre-X/IXV experimental vehicle, which

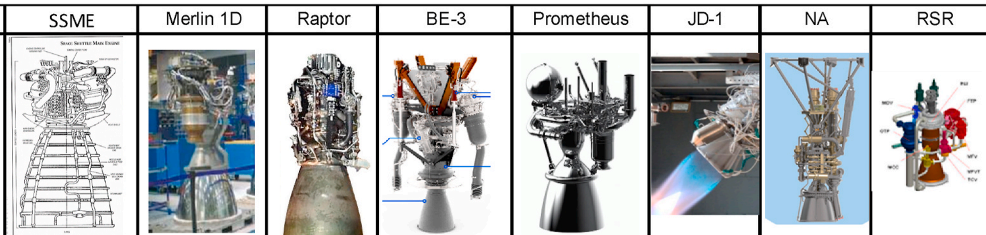
ENGINE	SSME	Merlin 1D	Raptor	BE-3	Prometheus	JD-1	NA	RSR
Image								
Vehicle	Space Shuttle	Falcon 9 v1.1 D	Starship	New Shepard	Ariane next gen	Hyperbola-2	SRV-1	Callisto
Cycle	SC	GG	SC	Tap-off	GG		GG	Exp bleed
Propellant	LOX/LH2	LOX/RP1	LOX/LCH4	LOX/LH2	LOX/LCH4	LOX/LCH4	LOX/LCH4	LOX/LH2
Thrust V (kN)	2279	981	2000	490 (See Level)	1000	150	100	44
Isp Vacuum (s)	452	311	380				285 (SL)	355
Chamber pressure (bar)	206	97	300		110			34
Throttling (%)	50-109				30-110		20-100	40-100
Company	Rocketdyne	SpaceX	SpaceX	Blue Origin	ArianeGroup	iSpace	LinkSpace	ISAS/JAXA

Fig. 28. Liquid reusable engines [49–51].

permitted to demonstrate technology maturity for re-entry TPS, aerodynamics and ATD, avionics, GNC for instance. But that is also true for a toss-back-barge-landing, fly-back booster in which system aspects are very interconnected.

Both for high or low energy vehicles (and for winged/unwinged boosters) the aerodynamic conception and characterization can be of major importance. However, the shape definition needs testing in low scale and this is mostly applicable for winged bodies.

For the in-flight demonstration and for the on-ground operations of the engine, the most interesting topics will be the load cases during ascent and re-entry phases and the fluid transient phases, which are difficult to be represented on ground tests. In-flight demonstrations can be useful to secure critical technology areas using eventually reduced scale vehicles. Furthermore, this demonstration can be fruitful to assess the actual capability to have limited turnaround and recovery costs to consolidate the interest of reusable booster stages depending on the operational vehicle constraints.

Finally, the main goal of the reusability demonstration is to explore part of typical flight envelope, test critical areas and technology, verify recovery & turnaround cost hypotheses. This is a system demonstration in the sense that the actual purpose is to test the complete system in a representative environment. Concerning the MRO (Maintenance, Repair and Overhaul), this is an essential part to assess through testing of ground and flight operations, vehicle maintenance, rapid turnaround and operational characteristics that will be relevant for future orbital vehicles and the associated recurring cost.

The compliance of the proposed demonstration activity has to be compared with this objective to assess the coverage of the objectives. This can be performed with one vehicle and more tests or different vehicles with a number of tests for each to be verified.

6. Summary and conclusion

This paper showed an overview of the reusable space systems with a view to high and low energy systems and the related technology aspects split between the two energy levels. Some year after the stop of space shuttle, reusability became relevant again for operational orbital spacecraft (both manned and unmanned) and for reusable boosters' stages. The former example is constituted by the winged/lifting body (X37-B, Dream Chaser, IXV/Space Rider …) and the partially reusable new generation re-entry capsules (Orion, Dragon …), the latter by the reusable toss-back/barge-landing boosters' stages developed by SpaceX and by Blue Origin (formerly for ballistic flights) and related demonstrators. The conical heat shields (including inflatable technology) can be a solution to recover and reuse upper stages.

The reusability of the booster stages can help reducing launch cost when performing missions with filling ratio less than 100 %, under specific production rate and refurbishment time/cost. The advantage depends strongly on the launch rate. Outside USA investigations to develop semi-reusable launchers are ongoing. Europe is currently developing the lifting body operational Space Rider in the ESA frame and has its demonstration program in the frame of future generations LVs (namely Callisto and Themis, the latter powered by Prometheus engine), but this is also the case of China and Japan.

However, even if the experience and feedback on semi-reusable rockets has grown in last years and seems a good boost for innovation, the development of new systems still needs critical technology maturation through subsystems and system tests, both on-ground and in-flight in order to secure operational flights. On its side France and Europe have developed and are developing reusable technology to enable future generation launch systems.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The author wishes to thank all CNES and ArianeGroup colleagues involved in current reusable systems developments and in particular Ch. Bonnal, S. Guedron for their precious help.

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